



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
Faculty of Science and Technology
Department of Computer Science
COE 3205 Computer Organization and Architecture (Section: J)

WORKSHEET 2F

QUESTION 1

1. **AND Operation:** 10101010 **AND** 11110000 → 1010 0000
2. **OR Operation:** 10101010 **OR** 11110000 → 1111 1010
3. **XOR Operation:** 10101010 **XOR** 11110000 → 0101 1010
4. **NOT Operation:** **NOT** 10101010 → 0101 0101

QUESTION 2

- Clear the sign bit of AL while leaving the other bits unchanged

AND AL, 7Fh

QUESTION 3

- Toggle the sign bit of DX.

XOR DX, 8000h

QUESTION 4

The ASCII codes range for

- "a" to "z" is 61h to 7Ah
- "A" to "Z" is 41h to 5Ah.

Convert a lowercase character stored in DL to UPPERCASE

Sub DL, 20h

QUESTION 5

Give an assembly code chunk to Jump to label BELOW If AL contains an even number.

TEST AL, 01H
JZ BELOW

QUESTION 6

Suppose DH contains 8Ah and CL contains 2. What are the values of DH and CF after the Instruction SHR DH, CL is executed?

STEP by STEP

- 10001010 → SHR 1 → 01000101 (CF = 0)
- 01000101 → SHR 1 → 00100010 (CF = 1)

Final Results:

- DH = 00100010b = 22h
- CF = 1 (last bit shifted out)



QUESTION 7

Use right shifts to divide the unsigned number 65143 by 4. Put the quotient in AX

```
MOV AX, 65143  
SHR AX, 2
```

QUESTION 8

Consider the following code:

```
MOV BX, 0xF218h.  
SHR BX, 3
```

What is the resulting value in BX, and what is the **Carry Flag (CF)** after the last bit is shifted out?

- BX = 1E43h
- CF = 1 (last bit shifted out)
- ✓ Zeros fill from the left
- ✓ Sign bit NOT preserved

Consider the following code:

```
MOV BX, 0xF218h.  
SAR BX, 3
```

What is the resulting value in BX, and what are the values of **SF**, **ZF**, and **PF** after the shift?

- BX = FE43h
- SF = 1 (sign bit = 1)
- ZF = 0 (result ≠ 0)
- PF = 1
- Low byte = 43h = 01000011b
- 4 ones → even parity

Explain the difference between *logical* and *arithmetic* right shift regarding how the **sign bit** is propagated.

Feature	SHR	SAR
Sign bit preserved?	No	Yes
Used for	Unsigned numbers	Signed numbers
Leftmost bit	Filled with 0	Filled with old sign bit